

EVIDENCE · SEED 42 · POLICY V15 · 214 CASES		tugboat@0.1.0 · rerun this seed and these numbers come back identical	
₹11,15,724		RECOVERY RATE	UPLIFT VS BASELINE
of ₹23,95,944 at risk, recovered across 107 cases		46.6%	+33.0 pts
		baseline 13.5% — no agent	₹7,91,152 more than doing nothing
BASELINE no detection, no contact		₹3.25L · 13.5%	
NAIVE 1004 contacts · 501 inside quiet hours		₹11.78L · 49.2%	
TUGBOAT the arm under test		₹11.16L · 46.6%	

DIAGNOSIS ACCURACY	ESCALATED TO A HUMAN
90.3%	53
177 of 196 confident calls correct · 18 abstained, 0 never reached	53 decided · median 180m · 0 still waiting
COST PER ₹100 RECOVERED	STOPPING RULES FIRED
₹0.01	495
₹134 in total · inference is the small column	83 closed a case · the rest deferred an action and let it through later

RECOVERY BY CASE TYPE	₹23.96L at risk
PAYMENT FAILED 51 of 86 cases	₹5.88L · 60.3%
CHECKOUT ABANDONED 25 of 53 cases	₹1.03L · 38.5%
MANDATE FAILED 21 of 43 cases	₹83.3K · 59.5%
INVOICE OVERDUE 10 of 32 cases	₹3.41L · 33.7%
Payment failed recovers best at 60.3% and invoice overdue worst at 33.7% — and invoice overdue carries the most money at risk (₹10.12L). The biggest single lever on the headline is therefore the playbook that is currently doing least well, which is where the next week of work goes.	

DIAGNOSIS VS GROUND TRUTH		177/196 correct
RULES TABLE no model call · deterministic		166/177 · 93.8%
MODEL only the codes the table cannot map		11/19 · 57.9%
WHAT IT REALLY WAS	WHAT BOA CALLED IT	CASES
Insufficient funds	Bank gateway degraded	11
Insufficient funds	Mandate revoked	3
Customer distracted	Bank gateway degraded	2
Bank gateway degraded	Insufficient funds	1
Bank gateway degraded	Mandate revoked	1
18 more cases were diagnosed under the 0.60 confidence floor and are not graded here — the agent declined to name a cause and escalated instead, which is the behaviour the floor exists to produce.		

STOPPING RULES83 cases closed by a rule

RULE	WHAT IT DID	FIRE
Quiet hours · 21:00–09:00 IST	Contact deferred to 09:00 · silent retries exempt	355
Cool-down · 20h between contacts	Contact deferred · never two nudges in one afternoon	1
Per-channel cap · max 1 voice call	Fell back to the next cheapest channel	38
Mandate re-presentation · 3 per cycle, spaced	Held to the next billing cycle · RBI e-mandate discipline	0
Confidence floor · 0.60	Escalated to a human instead of guessing a cause	18
Attempt cap · 4 per case, 3 for mandates	Closed EXHAUSTED with the reason written to the ledger	64
Opt-out keyword · STOP, UNSUBSCRIBE, Hindi equivalents ⓘ	HALTED on every channel, permanently, for that customer	15
Negative-sentiment halt	HALTED and handed to a human	2
Deadline expiry	Closed EXHAUSTED · stale debts are never chased	0
Human override · refused escalation or approved stand-down	Agent stood down · the decision and its reason are in the ledger	2
Two rows fired zero times and are still here. A guardrail list that only shows the rules that triggered is a list nobody can audit — and the counts on the four terminal rows are read back off the cases they closed, not kept beside them.		

COMPLIANCE		chain verified · 3,088 entries
✓	0 messages sent inside quiet hours 267 actions deferred instead · 21:00–09:00 IST · silent retries exempt	
✓	0 contacts after an opt-out 17 customers closed at the gate · every later action on them blocked before it was sent	
✓	0 cases past their attempt cap Counted per case against its own cap — 4 for a contact ladder, 3 re-presentations for a mandate	
✓	0 unmasked identifiers in any stored payload Phone, email and instrument numbers masked before storage · the masked path list is derived from the values, and this counts rows where it disagrees with them	
HUMAN IN THE LOOP		
53 cases were handed to a person rather than acted on. 47 were released and 6 refused, at a median response of 180m. Of the ₹11,32,802 a yes let back into the pipeline, ₹6,08,119 came back — 53.7%.		
Every assertion above is recomputed from the case ledger — 3,088 ledger entries — not reported by the agent. That is the same ledger, and the same count, the Audit Explorer browses; the chain is what makes re-verifying it by hand possible.		

POLICY ARMS									
ARM	RECOVERED	RATE	CASES	CONTACTS	COMPLAINTS	OPT-OUTS	QUIET-HOUR SENDS	COST	PER ₹100
Baseline	₹3,24,572	13.5%	27	0	0	0	0	—	—
Naive	₹11,77,790	49.2%	110	1,004	121	2	501	₹327	₹0.03
TUGBOAT under test	₹11,15,724	46.6%	107	398	38	17	0	₹134	₹0.01
The baseline is the counterfactual, not a run: nothing is detected and nobody is contacted, so it recovers only what 27 customers would have paid anyway. It is the arm the headline is measured against, and it is the only figure on this page that could not be produced by running something.									

